

Yoon-Ji Choi

M.S. Student in Mechanical Engineering
Interactive Robotics Lab, Dongguk University
Seoul, Republic of Korea

choiyj@dgu.ac.kr
github.com/Choiyoonji
choiyoonji.github.io

Research Interests

- Bimanual Robot Manipulation; Vision-Language-Action Models; Multimodal Perception; Humanoid Manipulation

Education

- M.S. in Mechanical Engineering**, Dongguk University, Seoul 2025–Present
Interactive Robotics Lab; Advisor: Prof. Soo-Chul Lim; GPA: 4.5/4.5; expected Aug. 2027.
- B.S. in Mechanical, Robotics and Energy Engineering**, Dongguk University, Seoul 2022–2025
GPA: 3.91/4.5.

Research Experience

- Graduate Researcher**, Interactive Robotics Lab, Dongguk University Sep. 2025–Present
- Developed a bimanual VLA architecture that couples task-aware multi-view selection with interaction-aware action decomposition; evaluated on six RoboTwin 2.0 tasks and three long-horizon real-robot tasks, improving average success by 27.7% in simulation and 43.3% on hardware over a monolithic baseline.
 - Built multimodal robot-learning pipelines using RGB, thermal, tactile, force, and proprioceptive signals; contributed to task-aware sensory routing, thermal-aware VLA, and interaction-force estimation research.
 - Integrated VR-based bimanual teleoperation, RBY1/UR5e control, dexterous-hand retargeting, simulation/real data collection, and VLA fine-tuning/inference pipelines using $\pi 0.5$ and GR00T.

Publications

Peer-Reviewed Journal Article

1. Young-Chae Son^{*}, Dae-Kwan Ko^{*}, **Yoon-Ji Choi**, and Soo-Chul Lim. 2026. “ThermoAct: Thermal-Aware Vision-Language-Action Models for Robotic Perception and Decision-Making.” *IEEE Robotics and Automation Letters*, 11(5):6106–6113.

Manuscripts Under Review

2. **Yoon-Ji Choi**, Young-Chae Son, and Soo-Chul Lim. 2026. “See Selectively, Act Adaptively: Dual-Level Structural Decomposition for Bimanual Robot Manipulation.” *IEEE Robotics and Automation Letters*, under review. arXiv:2606.13279.
3. Young-Chae Son^{*}, Jung-Woo Lee^{*}, **Yoon-Ji Choi**, Dae-Kwan Ko, and Soo-Chul Lim. 2026. “Task-Aware Modality Selection for Multimodal VLA in Robotic Manipulation.” *IEEE Robotics and Automation Letters*, under review.
4. Dae-Kwan Ko^{*}, Kang-il Yoon^{*}, **Yoon-Ji Choi**, Kang-Won Lee, and Soo-Chul Lim. 2026. “Vision-Language-Force Models as Physical Observers: Adaptation for Interaction Force Estimation.” *Information Fusion*, under review.

Selected Research Projects

- Data-Efficient Humanoid Autonomous Manipulation Intelligence** — National Research Foundation of Korea 2025–Present

- Built a VR-to-humanoid demonstration pipeline for RBY1 and a dexterous hand, including synchronized robot-state and sensor-data collection.
- Fine-tuned GR00T N1.7 with real and simulation-augmented demonstrations and integrated a distributed ROS2 inference/execution pipeline with a VLM planner and sub-task VLA servers.
- Performed real-robot evaluation and trajectory-level failure analysis to connect data composition, hardware behavior, and policy performance.

Language-Conditioned Robot Motion Generation with Visual/Tactile Multimodal Sensing — Ministry of Science and ICT

2025–Present

- Integrated thermal cameras and DIGIT tactile sensors for synchronized RGB, thermal, tactile, force, and robot-state data collection.
- Trained and evaluated multimodal VLA manipulation policies and implemented force-estimation baselines for physical-interaction analysis.

Patent

Co-inventor, “Robot Control Method, Apparatus, System, and Computer-Readable Recording Medium Using Thermal Information” 2026

Korean Patent Application No. 10-2026-0067117; filed Apr. 14, 2026.

Research Funding

AI Seoul Tech Research Support Program, Seoul Future Talent Foundation

2026

Selected researcher; one-year research support of KRW 20 million.

Selected Engineering Experience

Bimanual Teleoperation and Streaming IK — UR5e, RBY1, Inspire/RH56F1, Meta Quest, MuJoCo 2026

Integrated dual-arm teleoperation, geometry-based inter-arm collision avoidance, dexterous-hand retargeting, and a state-continuous StreamingIK solver with singularity damping and Cartesian/joint-space motion limits.

Mobile Manipulator Teleoperation and Data Collection System — Scout Mini, ROS, Python, React, WebSocket 2025

Built a distributed remote-control and data-collection framework with real-time camera/sensor streaming and an input-monitoring dashboard for a mobile manipulator.

SOOMAC Robot Arm Team, Founder and Team Lead

2024

Led an eight-member team developing collaborative-robot applications, including image-defined task execution and autonomous hamburger assembly; the team earned four competition awards in 2024.

Conference Presentation

“ThermoAct: Thermal-Aware Vision-Language-Action Models for Robotic Perception and Decision-Making” 2026
 Poster presentation, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026)*, Paper #238.

Honors and Awards

Chairman Award, Federation of Middle Market Enterprises of Korea (FOMEK), Creative Engineering Design Competition; **Best Award**, AIoT Creative Engineering Design Competition, Dongguk University 2025

Minister of Trade, Industry and Energy Award, Creative Engineering Design Competition; **Best Award**, AIoT Creative Engineering Design Competition, Dongguk University; **Director General Award (3rd Place)**, R-BIZ Challenge Zeus Robot Mission; **Encouragement Award**, Capstone Design Value-Up Program 2024

Innovation Award, Autonomous Driving S/W Advanced Competition, Seoul National University Future Mobility Technology Center; **Best Award**, AIoT Creative Engineering Design Competition, Dongguk University; **Excellence Award**, Capstone Design Value-Up Program; **Encouragement Award**, International Student EV Autonomous Driving Competition (AI Autonomous 1/5 Racer) 2023

Grand Prize, Future Mobility Urban Challenge; **Excellence Award**, Adventure Design Competition; **Excellence Award**, IoT Home Appliance Consortium Universities Creative Engineering Design Competition; **Award**, Creative Engineering Design Competition, Korean Society for Engineering Education 2022

Technical Skills

Programming: Python, C/C++, JavaScript, Node.js

AI / ML: PyTorch, scikit-learn, Vision-Language-Action models, multimodal learning, computer vision

Robotics: ROS / ROS 2, MuJoCo, Isaac Sim, OpenCV, teleoperation, inverse kinematics

VLA / Models: π 0.5, GR00T N1.7

Platforms: UR5e, RBY1, Inspire/RH56F1, Kinova Gen3 Lite, Scout Mini, Meta Quest, Jetson, Raspberry Pi

Languages: Korean (native); English (OPIC Intermediate High, Sep. 2026)